

E090

Applied AI Engineering

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

The vice president at a streaming company, who just returned from a conference in Lisbon, tells the capstone team to “use AI to reduce churn.” The company has 4 million subscribers. What should the team do first?

- A. Collect every data source the company has.
- B. Pick the most advanced deep learning model available.
- C. Use accuracy, which is always the right success metric.
- D. Define the churn target and time window, the action it will drive, and a measurable success metric.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A capstone team’s claim-routing model reaches 88% accuracy, well above an 80% figure quoted in an industry blog. The insurer’s current rule-based system routes 90% of claims correctly, and 85% of claims go to the most common queue. Is the model ready to replace the rules?

- A. Yes; it beats the 80% industry figure.
- B. No; it does not beat the existing 90% rule system.
- C. Yes; it beats the 85% majority-class rate.
- D. Yes; machine learning always outperforms rules.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

In a dataset, 92% of transactions are legitimate. What accuracy does a trivial model reach by always predicting “legitimate”?

- A. 92%
- B. 8%
- C. 50%
- D. 0%, because it never finds fraud.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A clinic-scheduling capstone compares two no-show models on the same month. Model A has 40 false negatives and 200 false positives; Model B has 20 false negatives and 600 false positives. A missed no-show costs \$500 and an unneeded reminder call costs \$20. Model B's dashboard uses nicer charts. Which model has lower total error cost?

- A. Model A, because it makes fewer total errors
- B. Model A, at \$24,000
- C. Model B, at \$22,000
- D. Model B, at \$10,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Model A has 40 false negatives and 200 false positives; Model B has 20 and 600. False positives cost \$20 each. At what false-negative cost would the two models cost the same?

- A. \$200
- B. \$400
- C. \$8,000
- D. \$20

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A capstone team at a startup with a dog-friendly office builds its API container once for staging, tests it there, and then rebuilds a new image from the same repo for production. The pipeline runs on a hosted CI service. What is the problem and fix?

- A. No problem; containers behave identically however they are built.
- B. Rebuild the image in every environment for freshness.
- C. Build once, push to a registry, and promote the same tested image to production.
- D. Skip staging tests to deploy faster.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A bank's new fraud model must be compared with the current model on live transactions without affecting any customer. The bank's app recently won a design award, and its data center is in Ohio. Which deployment strategy fits?

- A. Shadow deployment
- B. Replace the old model immediately
- C. Blue-green switch
- D. Canary release

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A new model serves 5% of real users while error rates are watched; the share grows if all goes well. What is this strategy?

- A. Blue-green deployment
- B. Canary release
- C. Big-bang release
- D. Shadow deployment

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A loan-default model at a lender with branches along the coast shows stable input distributions month to month, but its precision has fallen sharply since interest rates changed. The model runs on a Kubernetes cluster. What is happening?

- A. Concept drift; the link between inputs and default has changed.
- B. Data drift; the inputs have shifted.
- C. Nothing; stable inputs mean the model is still fine.
- D. A Kubernetes failure.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Monitoring shows the median applicant income has risen 30% since training, but outcome labels will not be available for months. What should the team conclude and do?

- A.** Ignore it until labels arrive.
- B.** Retrain immediately; any drift always requires it.
- C.** Concept drift has been proven.
- D.** Data drift; alert and investigate its effect before labels arrive.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An auditor at a health-tech company with a new downtown lab asks which dataset, code, and settings produced the model that ran in production last March. The team has only a file named model_final_v2.pkl. The company uses cloud storage. What would have made this answerable?

- A.** Clearer file names such as model_final_v3.pkl.
- B.** Saving the weights file more often.
- C.** Committing the large training data to Git alone.
- D.** A model registry linking each version to its data version, code commit, hyperparameters, and metrics.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A capstone team's rental-application screening system, built with a popular open-source library, has an AUC of 0.91. The landlord client owns 12 buildings. What must be done before launch?

- A.** Remove human review so decisions are consistent.
- B.** Add safeguards only if complaints arrive.
- C.** Check error rates across groups, add human review for denials, document limits, and review legal requirements.
- D.** Launch; an AUC of 0.91 is enough.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

The vice president at a streaming company, ~~who just returned from a conference in Lisbon~~, tells the capstone team to “use AI to reduce churn.” ~~The company has 4 million subscribers.~~ What should the team do first?

Target word: first lifecycle step **Decides:** The goal is not yet a defined ML problem.

A	Adjacent category	Data needs depend on the target and decision.
B	Inverted operation	Model choice follows problem definition, not the other way around.
C	Absolute language	The metric must fit the business cost of errors.
D	Correct	Formulation turns a vague goal into something testable.

2. Answer B Concept 2

A capstone team’s claim-routing model reaches 88% accuracy, ~~well above an 80% figure quoted in an industry blog~~. The insurer’s current rule-based system routes 90% of claims correctly, and 85% of claims go to the most common queue. Is the model ready to replace the rules?

Target word: compare against the right baseline **Decides:** The existing system scores 90%.

A	Camouflage	An outside number does not decide this replacement.
B	Correct	The relevant baseline is what it would replace.
C	Dropped element	Ignores the stronger rule-based baseline.
D	Absolute language	The data here show the rules performing better.

3. Answer A Concept 2 · transfer

In a dataset, 92% of transactions are legitimate. What accuracy does a trivial model reach by always predicting “legitimate”?

Target word: solve for the trivial baseline **Decides:** Majority-class share.

A	Correct	It is right on every legitimate transaction.
B	Inverted operation	That is the minority share.
C	Adjacent category	A coin-flip rate, not the majority-class rate.
D	Confusion set	Mixes up fraud recall (0) with overall accuracy.

4. Answer C Concept 3

A clinic-scheduling capstone compares two no-show models on the same month. Model A has 40 false negatives and 200 false positives; Model B has 20 false negatives and 600 false positives. A missed no-show costs \$500 and an unneeded reminder call costs \$20. ~~Model B’s dashboard uses nicer charts.~~ Which model has lower total error cost?

Target word: total cost comparison **Decides:** $A = 40(500) + 200(20) = \$24,000$; $B = 20(500) + 600(20) = \$22,000$.

A	Adjacent category	240 vs. 620 errors ignores the unequal costs.
B	Inverted operation	Picks the higher-cost model.
C	Correct	$\$10,000 + \$12,000 = \$22,000$, less than A’s \$24,000.
D	Dropped element	Leaves out the false-positive costs.

5. Answer B Concept 3 · transfer

Model A has 40 false negatives and 200 false positives; Model B has 20 and 600. False positives cost \$20 each. At what false-negative cost would the two models cost the same?

Target word: solve for break-even FN cost **Decides:** $40c + 4,000 = 20c + 12,000 \rightarrow 20c = 8,000$.

A	Adjacent category	Divides by 40, A's FN count, instead of the difference.
B	Correct	$8,000 \div 20 = 400$.
C	Dropped element	Forgets to divide by the 20-case FN difference.
D	Wrong stem number	The false-positive cost given in the stem, not the break-even FN cost.

6. Answer C Concept 4

A capstone team at a startup ~~with a dog-friendly office~~ builds its API container once for staging, tests it there, and then rebuilds a new image from the same repo for production. ~~The pipeline runs on a hosted CI service.~~ What is the problem and fix?

Target word: promotion practice **Decides:** Production runs an image that was never tested.

A	Absolute language	Rebuilds can pick up different base images or packages.
B	Inverted operation	Each rebuild can pull different dependencies.
C	Correct	What was tested is exactly what ships.
D	Dropped element	Removes the check that catches failures.

7. Answer A Concept 5

A bank's new fraud model must be compared with the current model on live transactions without affecting any customer. ~~The bank's app recently won a design award, and its data center is in Ohio.~~ Which deployment strategy fits?

Target word: zero-impact live comparison **Decides:** Live traffic, but no customer impact.

A	Correct	The new model scores live traffic, but only the current model's decisions are used.
B	Inverted operation	Exposes every customer to an unproven model.
C	Confusion set	Moves all traffic to the new model at once.
D	Adjacent category	A small share of real customers is affected.

8. Answer B Concept 5 · transfer

A new model serves 5% of real users while error rates are watched; the share grows if all goes well. What is this strategy?

Target word: reverse: description to strategy **Decides:** A small share of real users first.

A	Adjacent category	Switches all traffic between two environments.
B	Correct	Gradual exposure of real users with monitoring.
C	Inverted operation	Sends everyone to the new model at once.
D	Confusion set	Shadow outputs never affect users.

9. Answer A Concept 6

A loan-default model at a lender ~~with branches along the coast~~ shows stable input distributions month to month, but its precision has fallen sharply since interest rates changed. ~~The model runs on a Kubernetes cluster.~~ What is happening?

Target word: drift type **Decides:** Inputs stable, performance down.

A	Correct	Same kinds of inputs, different outcomes.
B	Confusion set	The stem says input distributions are stable.
C	Dropped element	Ignores the drop in precision.
D	Camouflage	Infrastructure does not explain lower precision on stable inputs.

10. Answer D Concept 6 · transfer

Monitoring shows the median applicant income has risen 30% since training, but outcome labels will not be available for months. What should the team conclude and do?

Target word: reverse: inputs changed, labels unknown **Decides:** An input distribution shifted.

A	Dropped element	Wastes the early warning.
B	Absolute language	Investigate impact first; not all drift hurts performance.
C	Inverted operation	Concept drift needs outcome labels to confirm.
D	Correct	Input shift is visible now and may hurt performance.

11. Answer D Concept 7

An auditor at a health-tech company ~~with a new downtown lab~~ asks which dataset, code, and settings produced the model that ran in production last March. The team has only a file named model_final_v2.pkl. ~~The company uses cloud storage.~~ What would have made this answerable?

Target word: reproducibility control **Decides:** No link between model and its inputs.

A	Adjacent category	Names are not a trustworthy record.
B	Dropped element	Weights alone lack data and code lineage.
C	Confusion set	Plain Git handles large data poorly and still misses run settings and metrics.
D	Correct	Records full lineage for every deployed model.

12. Answer C Concept 8

A capstone team's rental-application screening system, ~~built with a popular open-source library,~~ has an AUC of 0.91. ~~The landlord client owns 12 buildings.~~ What must be done before launch?

Target word: responsible launch steps **Decides:** Automated decisions affect people's access to housing.

A	Inverted operation	Removes the safeguard for adverse outcomes.
B	Absolute language	Responsible practice builds safeguards in before harm occurs.
C	Correct	High-impact decisions need fairness checks, oversight, and compliance.
D	Dropped element	Overall metrics can hide group harms.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Problem formulation	Before modeling, define the prediction target, the decision it drives, a measurable success metric, and constraints.	Starting with the most advanced model before the problem is defined.	A vague goal such as “use AI to reduce churn.”
2	Baselines	A model is useful only if it beats the simplest reasonable baseline — the majority class or the existing rule system.	Judging a score in isolation or against an unrelated benchmark.	A model score with a class balance or an existing system’s score nearby.
3	Cost-weighted evaluation	When error types cost different amounts, compare models by total expected cost ($FN \times \text{cost_FN} + FP \times \text{cost_FP}$), not by error counts.	Picking the model with fewer total errors.	Dollar costs attached to false positives and false negatives.
4	CI/CD with containers	Test, build the image once, push it to a registry, and promote the same tested image through staging to production.	Rebuilding separately for each environment or skipping tests to ship faster.	A pipeline that builds more than once or deploys untested code.
5	Deployment strategies	Shadow = new model sees live traffic but its outputs are not used; canary = a small share of real users first; blue-green = switch all traffic between two environments with fast rollback.	Calling any gradual rollout “shadow,” or exposing users when the goal was zero impact.	How much real traffic the new model’s decisions affect.
6	Data drift vs. concept drift	Data drift = input distributions change; concept drift = the input-to-outcome relationship changes, found by monitoring performance against labels.	Assuming stable inputs mean a stable model.	Which changed: the inputs, or the accuracy on the same kind of inputs.
7	Versioning and reproducibility	A model registry and experiment tracking tie each model version to its data version, code commit, hyperparameters, and metrics.	Saving only the weights file or relying on file names.	An audit question about what produced a model.
8	Responsible AI across the lifecycle	High-impact systems need subgroup fairness checks, human review for adverse outcomes, documentation, and legal review before launch.	Treating strong overall metrics as sufficient, or adding safeguards only after complaints.	Automated decisions about people’s access to jobs, housing, credit, or services.

Course scope (public description): This capstone course requires students to integrate and apply engineering knowledge from mathematics, programming, data engineering, machine learning, deep learning, and software engineering to design, validate, and deploy an AI-enabled system that addresses a real-world problem. Learners work through the full AI system lifecycle, including problem formulation, model development, testing, deployment, and monitoring. The course emphasizes ethical and responsible AI practices, systematic evaluation using appropriate metrics, and production deployment using industry-standard MLOps tools such as cloud infrastructure, containerization, CI/CD pipelines, and monitoring frameworks. Through this culminating experience, learners demonstrate the ability to design and deploy AI solutions that meet technical, ethical, and operational constraints expected of entry-level AI engineers.